Welcome to Code Crunch!

C1|DNA Strand Synthesis

[Fri 2-3PM] Weekly | Spring 2025

Attendance

linktr.ee/CODE.CRUNCH

Unit #2

Understanding the Function of DNA, and how to copy it

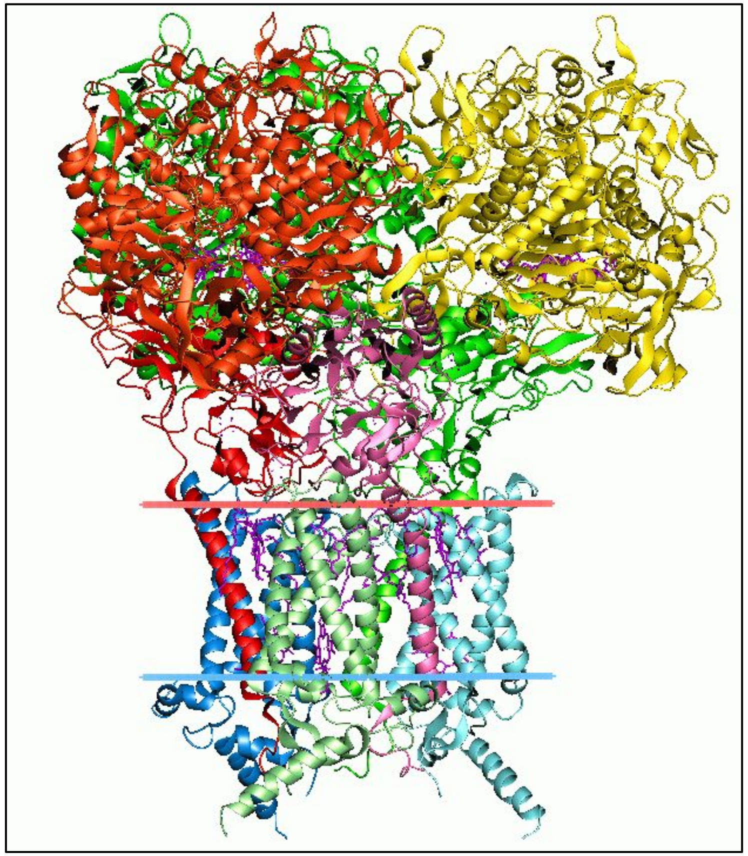
Previously...

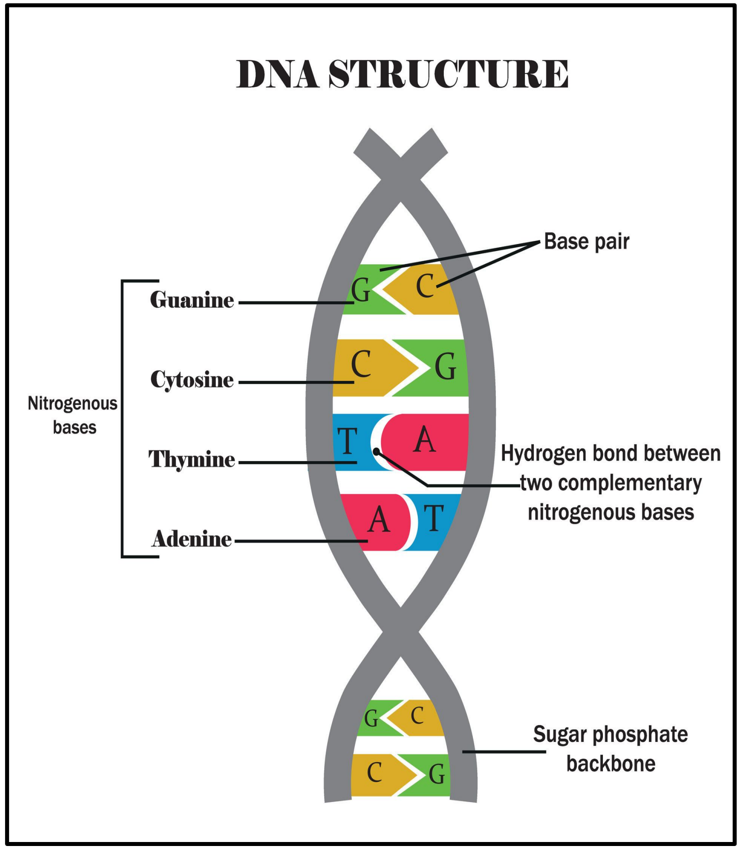
Introduction to Bioinformatics

Applications: Health & Research

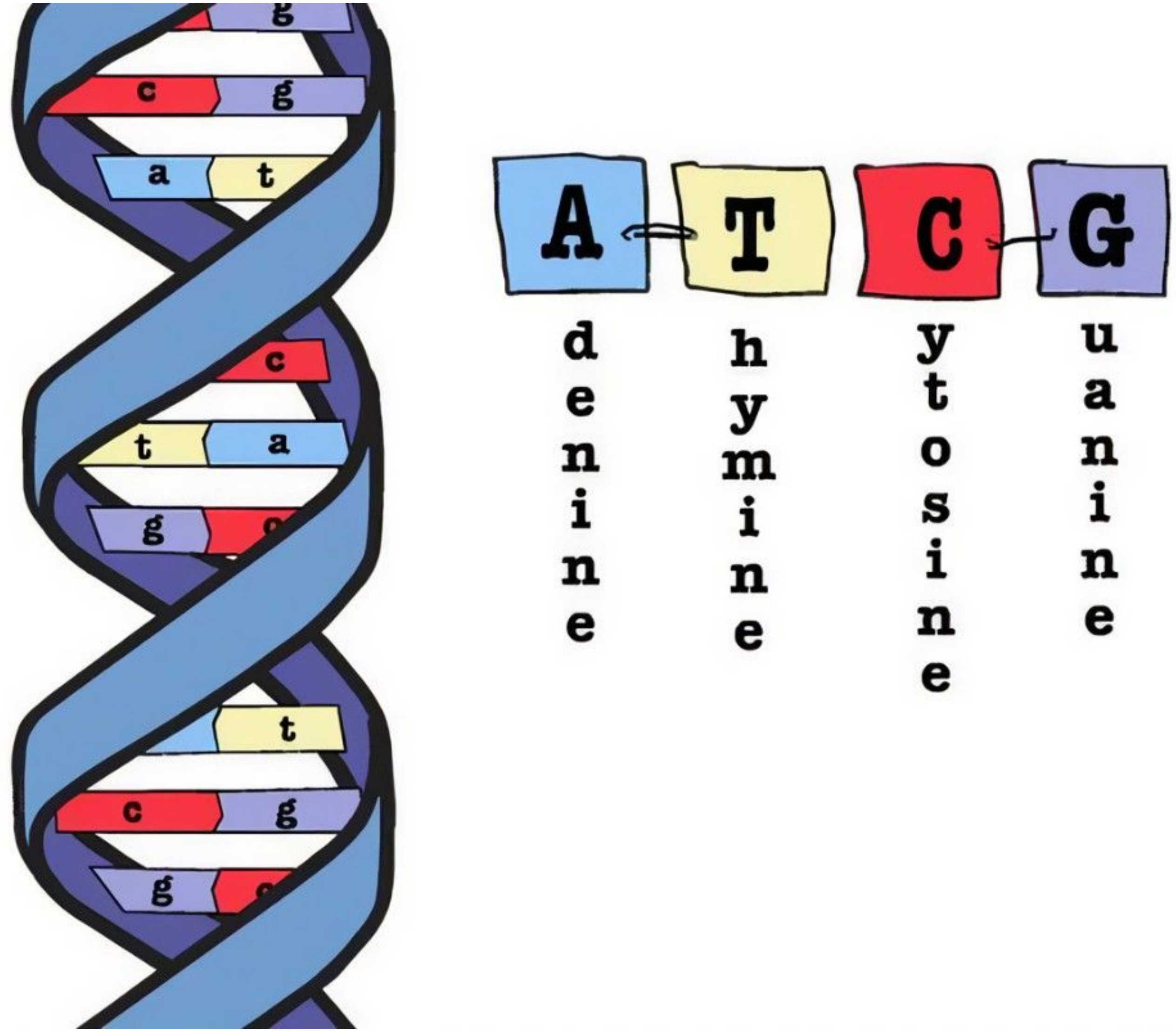
DNA_Strand Test Function

Biological Data Types:
- DNA
- RNA
- Proteins





DNA Base Reference Formula:



A <-> T

C <-> G

DNA_STRAND TEST Function

We tested if a certain strand is DNA Compatible.

Output:
False

```
dna_seq = "AGTJCTCTCTCCA" #Our DNA Strand

DNANucleotides = ['A', 'C','G','T']

# Function 1: Validating a DNA sequence if its true
def ValidateSeq(dna_seq):
tmpseq = dna_seq.upper()
 for nuc in tmpseq:
    if nuc not in DNANucleotides:
          return False
 return tmpseq

ValidateSeq(dna_seq)
```

Input your own DNA_Strand to Test

```
dna_seq = input("Type your DNA Strand here:")


# ------------------------------------------------------ FUNCTION 1 VALIDATION OF
SEQUENCE --------------------------------
DNANucleotides = ['A', 'C','G','T']
# Function 1: Validating a sequence if its true
def ValidateSeq(dna_seq):
  tmpseq = dna_seq.upper()
  for nuc in tmpseq:
      if nuc not in DNANucleotides:
        return print("Sorry that's not a DNA Strand :(")
  return tmpseq



print("Yay, this is a DNA Strand:" + ValidateSeq(dna_seq))
```

Random_DNA Generator Function

- We can make code, randomly generate us DNA strand and we can check if its a real DNA with ValidateSeq() Function

```
# Given we have ValidateSequence Function and List of Nucelotides
import random
randDNAstr = ''.join([random.choice(DNANucleotides) for nuc in range(20)])
print("Here is a randomly generated DNA: " + ValidateSeq(randDNAstr))
```

Random_DNA Generator Full Code

```
DNANucleotides = ['A', 'C','G','T']
# Function 1: Validating a sequence if its true
def ValidateSeq(dna_seq):
  tmpseq = dna_seq.upper()
  for nuc in tmpseq:
      if nuc not in DNANucleotides:
        return print("Sorry that's not a DNA Strand :(")
        break
  return tmpseq

import random
randDNAstr = ''.join([random.choice(DNANucleotides) for nuc in range(20)])
print("Here is a randomly generated DNA: " + ValidateSeq(randDNAstr))
DNAstring = ValidateSeq(randDNAstr)
```

Counting_Nucleotides Challenge

You are told by your Research Laboratory Instructor Dr. Ash to make a Random DNA strand and use it to count the number of nucleotides in it. Count how many Guanidines, Adenines, Cytosines, and Thymines it has? In other words, all the A,T,C,Gs in a randomly generate DNA strand.

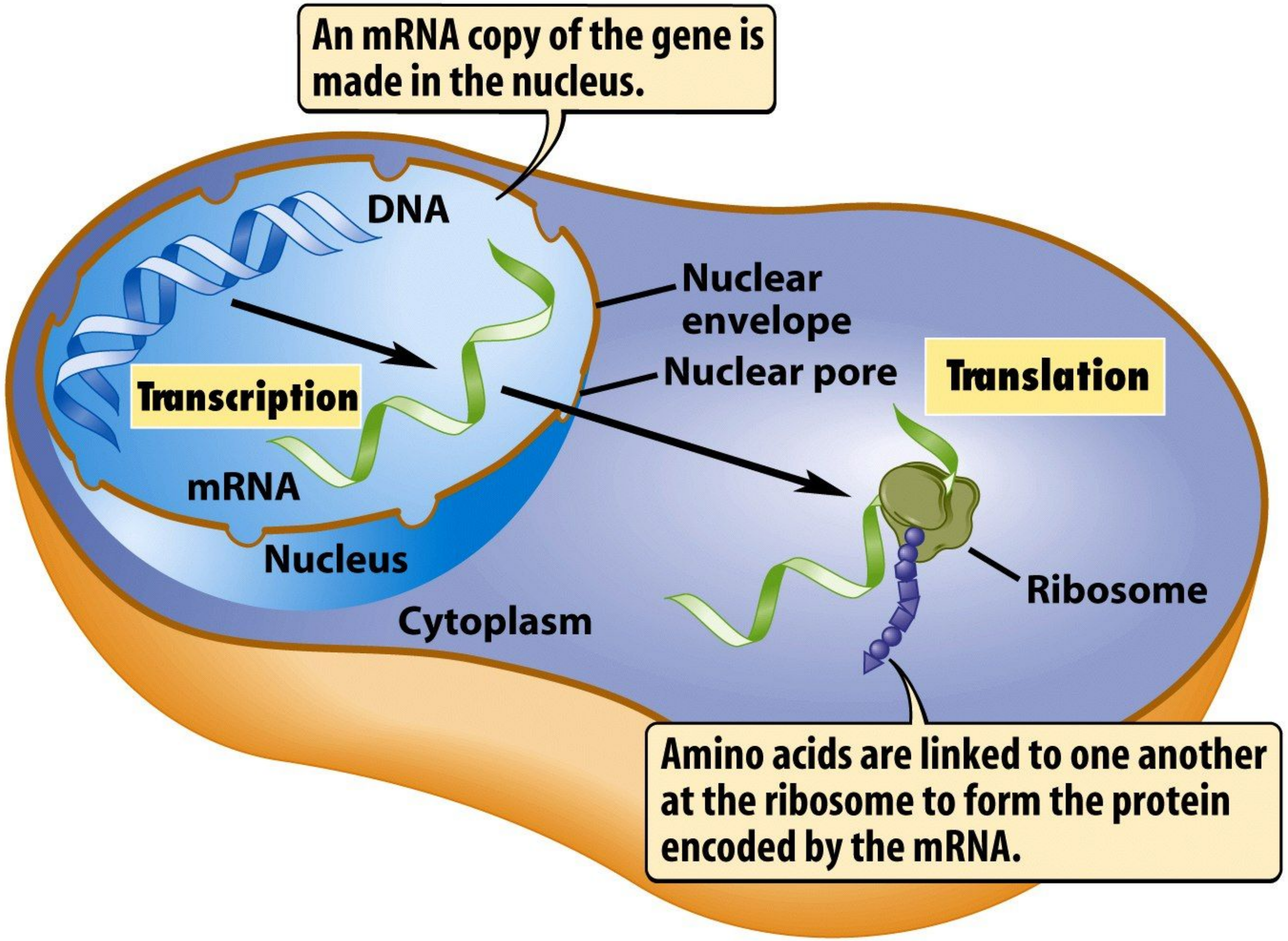
Counting_Nucleotide() Function

```python
# Function 3 Counting Nucleotides
def countNucfrequency(seq):
  temporaryFrequencyDictionary = {"A": 0, "C": 0, "G": 0, "T": 0}
  for nuc in DNAstring:
    temporaryFrequencyDictionary[nuc] += 1
  return temporaryFrequencyDictionary


print(countNucfrequency(DNAstring))
```

Counting_Nucleotides Full Work

```
DNANucleotides = ['A', 'C','G','T']
import random

def ValidateSeq(dna_seq): #Function 1
  tmpseq = dna_seq.upper()
  for nuc in tmpseq:
      if nuc not in DNANucleotides:
        return print("Sorry that's not a DNA Strand :(")
        break
  return tmpseq

randDNAstr = ''.join([random.choice(DNANucleotides) for nuc in range(20)]) # Function 2
DNAstring = ValidateSeq(randDNAstr)

def countNucfrequency(seq):   #Function 3
  temporaryFrequencyDictionary = {"A": 0, "C": 0, "G": 0, "T": 0}
  for nuc in DNAstring:
    temporaryFrequencyDictionary[nuc] += 1
  return temporaryFrequencyDictionary

print("DNA Strand:" + ValidateSeq(randDNAstr))
print(countNucfrequency(DNAstring))
```

Biological Concept - DNA Copying Transcription & Translation



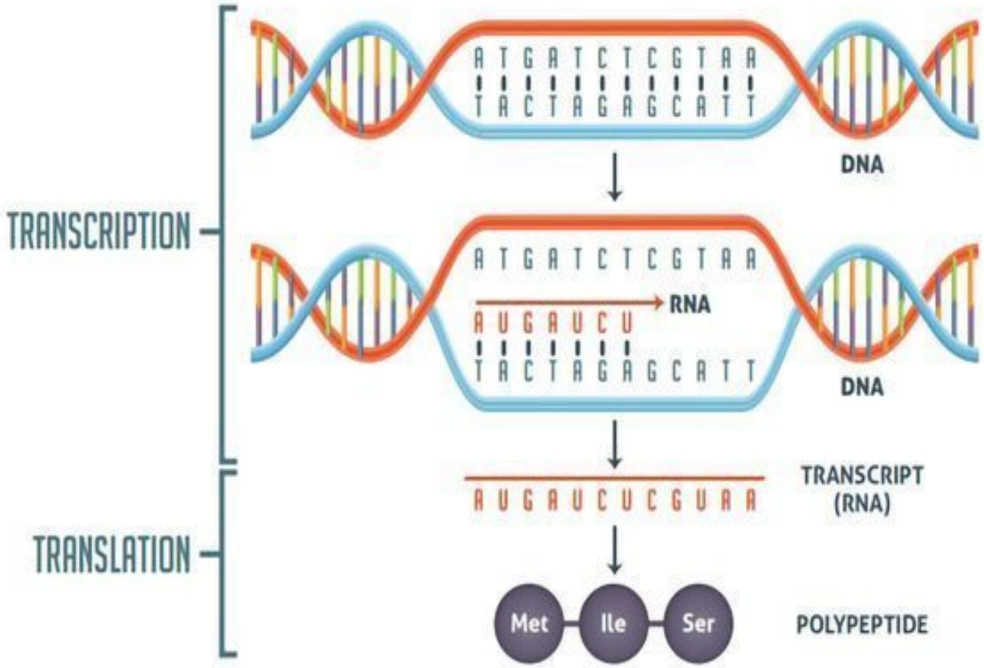


Figure 13-2 Discover Biology 3/e

Transcription (in Python)

```
def transcription(seq):
  return seq.replace("T", "U")


print(transcription(DNAstring))
```

Challenge 2: DNA -> RNA

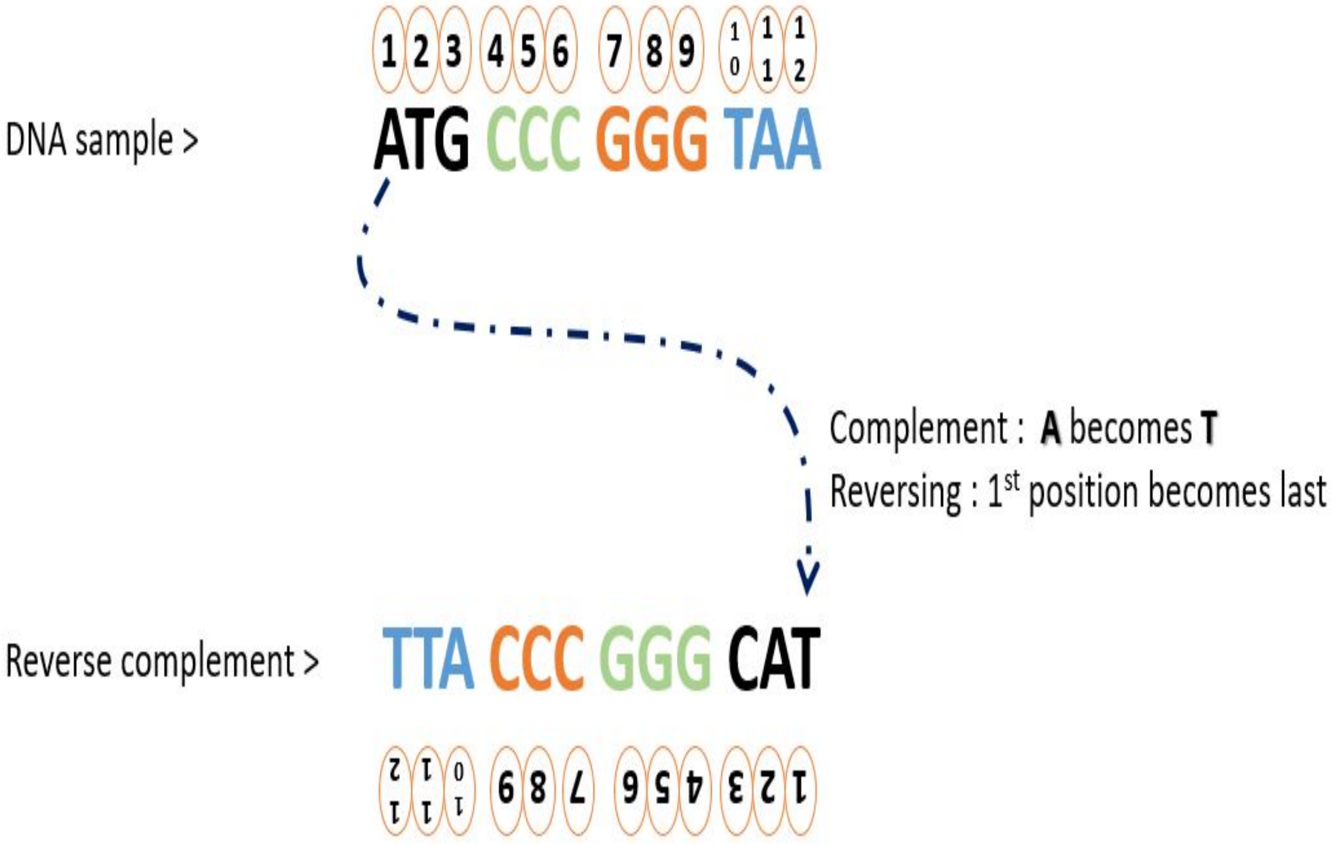
Dr. Ash was pleased you were able to count DNA bases from a random string of DNA. Now he wants you to replicate a messenger RNA from DNA via Transcription. How can you turn your DNA strand into RNA?

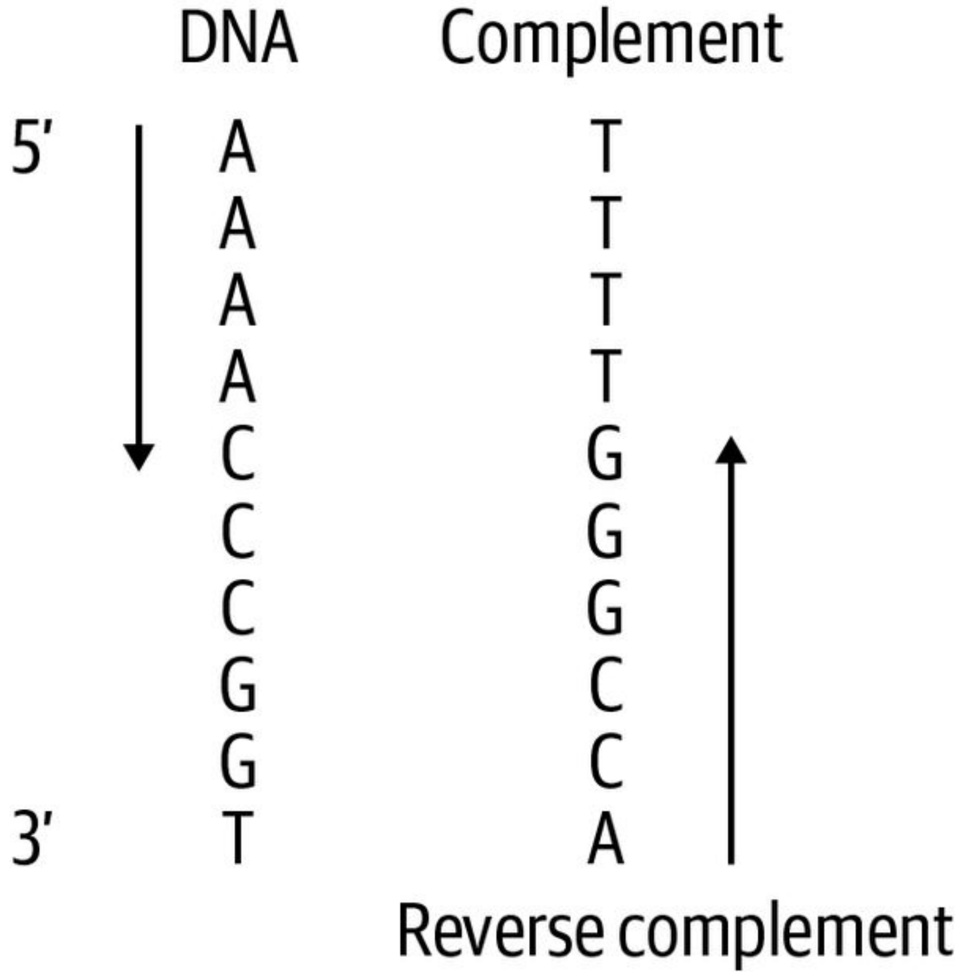
(Hint: Search the differences in Bases in DNA and RNA)

Transcription (in python)

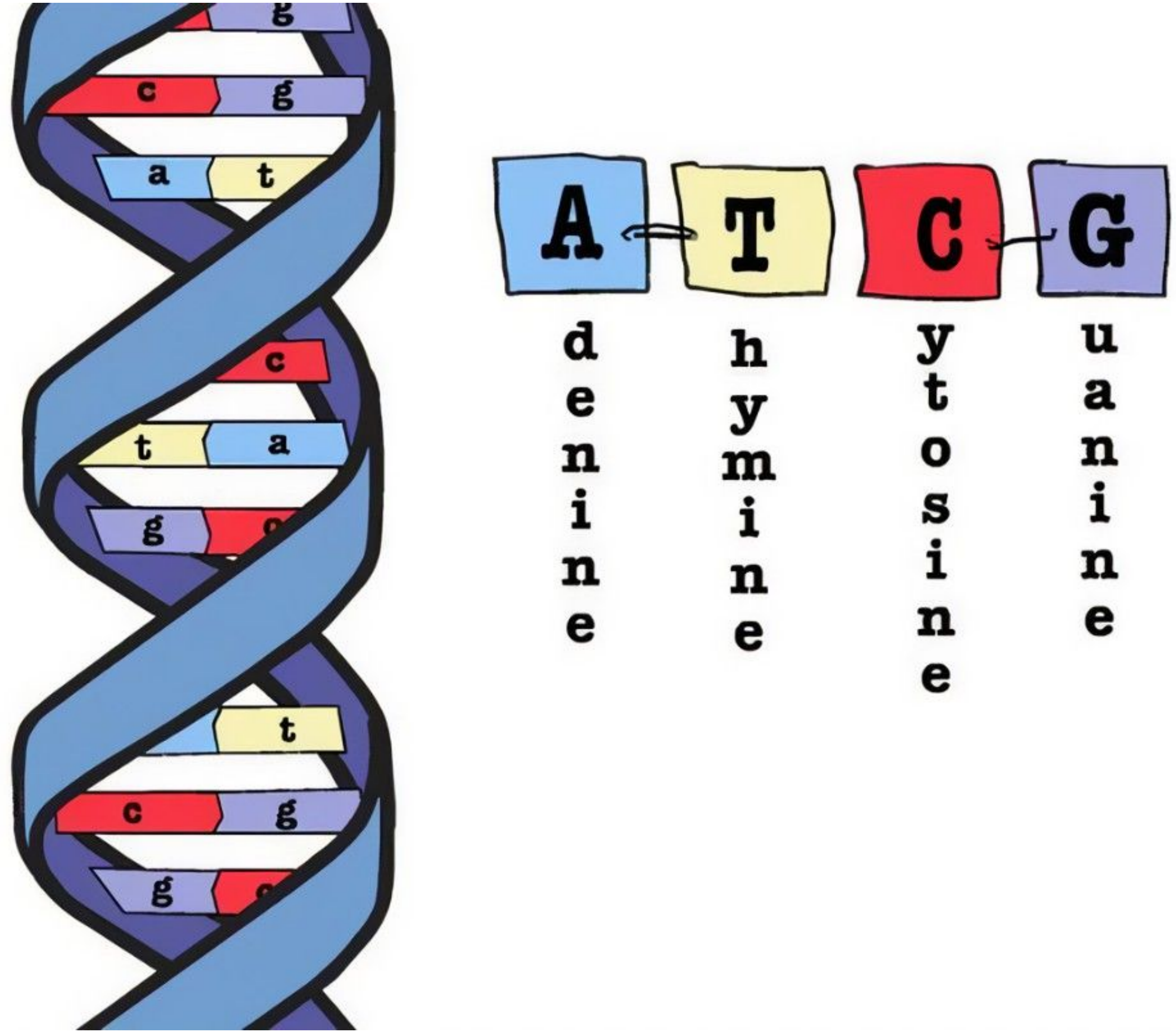
```
import random
#FUNCTION2:
randDNAstr = ''.join([random.choice(DNANucleotides) for nuc in range(20)])
print("Here is a randomly generated DNA: " + ValidateSeq(randDNAstr))
DNAstring = ValidateSeq(randDNAstr)

#FUNCTION 4:
def transcription(seq):
  return seq.replace("T", "U")
print("DNA Strand:" + ValidateSeq(randDNAstr)) # Makes sure we use DNA
print(countNucfrequency(DNAstring)) # Counts Bases
print(transcription(DNAstring)) # Turned DNA -> RNA
```

Reverse Complement in Bioinformatics





DNA Base Reference Formula:



A <-> T

C <-> G

5’ end **A C T G** *3’ end*

reverse complement

5’ end **C A G T** *3’ end*

Q&A Session

Attendance

Join us for the upcoming weekly sessions this Spring 2025!

Monday: C5 1PM / **C3** 2PM
Tuesdays: C7 1PM
Wednesdays: C5 1PM / **C1** 2PM
Thursdays: C6 1PM / **C4** 2PM / **C8** 3PM
Friday: C2 1PM & **C9** 2PM

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